CRUST

David Skocic

February 17, 2026

MTH 496H - Honors Senior Project

Fall Semester 2025

Advisor: Dr. Gregory Lupton

Cleveland State University

Department of Mathematics & Statistics

Contents

Abstract

...

1 Introduction and Background

Describe the analysis of algorithms

Define curve reconstruction

give examples why connecting points to the nearest neighbor isn't good

Maybe define "general position"

The crust method of curve reconstruction builds on a few basic geometric tools that are used constantly in discrete geometry in a wide degree of applications. The following sections introduce these tools and prove numerous important and interesting results about them.

2 Convex Hulls

The first geometric tool of interest is the convex hull. A convex hull is used to take a set of points S and connect them to make a convex shape that contains every point in S. Here, containing a point means that it is either be inside the shape or on its edge. It is easy to imagine that there are all sorts of ways to contain every point in S with some sort of convex polygon. You could, for example, draw a square so large that it contains every point in S, but a shape unrelated to how the points are arranged in S provides almost no more utility than just knowing S. A convex hull fixes this problem, and adheres to the following definition:

Definition 1 (Convex Hull)**.** *For a given set of points S, the convex hull is the intersection of every convex shape that contains every point in S.*

SHOW THAT THE INTERSECTION OF TWO CONVEX SHAPES IS ALSO CONVEX

Imagining such a shape for a given set of points is easy. To generate the convex hull for a set of points, imagine the points are nails in a wall. Start by tying a string to a nail at some extrema, say, the leftmost nail on the wall, and holding the string left of that. Then, rotate it all the way around all the other nails until it touches the first one again. This shape the string takes on is the convex hull.

The convex hull is a convenient tool because it has an intuitive definition and because convex shapes tend to be simple to work with. Also, it can serve as a starting point in an algorithm that starts with an unlabeled set of points like the crust. Taking the convex hull at least gives us a place to start connecting our points, even if its usefulness may not initially be clear.

Fortunately, lots of time has been spent finding optimal algorithms for computing the convex hull of a set of points. Computing the convex hull really means to find which points from S are vertices of the hull. This can be done in a variety of ways. One being a sort of inductive method that starts with 3 points, then add one at a time from the set, removing points that end up making the shape nonconvex. Another algorithm

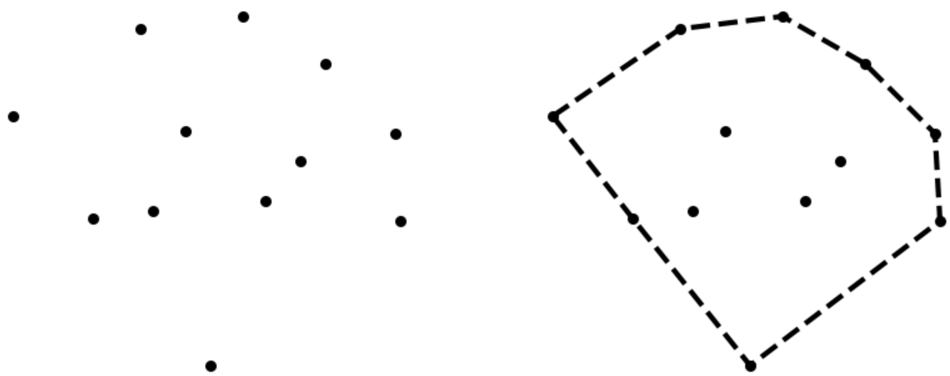


Figure 1: A set of points next to their convex hull

behaves very similarly to the string wrapping algorithm. While these algorithms are both intuitive, there is another class of algorithms that take less time to compute.

There are numerous algorithms that belong to this optimal class, all with time complexity $O(n \log n)$. From these I will focus on one which uses a divide and conquer approach. I choose this method because of its satisfying recursive nature, and because it is the algorithm that most easily extends to a third dimension. My implementation of this algorithm is largely inspired by the description from Devadoss and O'Rourke.

The pseudocode for this algorithm is shown in Algorithm 1. But it begins by taking in a list of coordinates sorted by their x-value. Since the algorithm is recursive it must have a base case, and here that base case is if there are three or fewer points in S because a triangle is already a convex polygon. When there are more than three points in S, separate the points into the left half and the right half and generate the convex hull of those points. This splitting continues until the hulls are triangles or line segments, which is the base case. The magic of the algorithm comes when bringing the groups back together. To do this there are two convex hulls, one on the right and one on the left. To combine them, a tangent line is found on the top and bottom (need a figure for this). *continue this description*

3 Triangulations

Triangulations are a key tool in the process of curve reconstruction. Specifically, a type of triangulation called a Delaunay triangulation is used to construct the crust of a given set of points. Before diving into Delaunay triangulations and some interesting results that follow, we first explore an easier problem, the

Algorithm 1 Convex Hull

Require: S, a list of points sorted by their x-coordinate
  function ConvexHull(S)
    if $|S| \leq 4$ **then**
      return S
    else
      $L \leftarrow \textsc{ConvexHull}(S[0 \dots \lfloor\frac{|S|}{2}\rfloor])$
      $R \leftarrow \textsc{ConvexHull}(S[\lfloor\frac{|S|}{2}\rfloor \dots |S|-1])$
      return Combine(L, R)
    end if
  end function
  function Combine(L,R)
    Fill this in
  end function

triangulation of a polygon, and build up to Delaunay triangulations from there.

3.1 Triangulations of Polygons

To begin, we must first define a triangulation of a polygon.

Definition 2 (Triangulation of a Polygon)**.** *A decomposition of a polygon into triangles by a maximal set of non-intersecting diagonals.*

In simpler terms, a triangulation breaks a polygon into triangular pieces by drawing segments between its vertices so that no two segments intersect. The term 'maximal set' is used in the definition to ensure that there is no vertex of the polygon on the edge of any triangle.

As an example, let's look at an easy case: a convex octagon. It is easy to pick out a way to triangulate this shape: just start at one vertex and draw lines to every other vertex as shown in Figure 2. It is easy to see how this technique can extend to any convex n-gon to create a triangulation with n-2 triangles. Something of note is that there can be more than one way to triangulate a polygon, with another example shown in Figure 2. Is it possible that some triangulations are better than others? The fact that there is a section dedicated to a specific type, the Delaunay triangulation, should be a clue here, and the reason why is explored in that section.

A more interesting problem arises when you try to triangulate a shape that is nonconvex. Is this even possible for *any* polygon? If it is, is there an algorithmic way that works to triangulate any polygon? Will there always be the same number of triangles? The following theorems dive deeper into these questions.

Theorem 1. *All polygons can be triangulated*

Proof. This is an inductive proof with the base case being a polygon with 3 vertices. This base case is trivial because a triangle need not be broken down into more triangular pieces. Now, the assumption that any

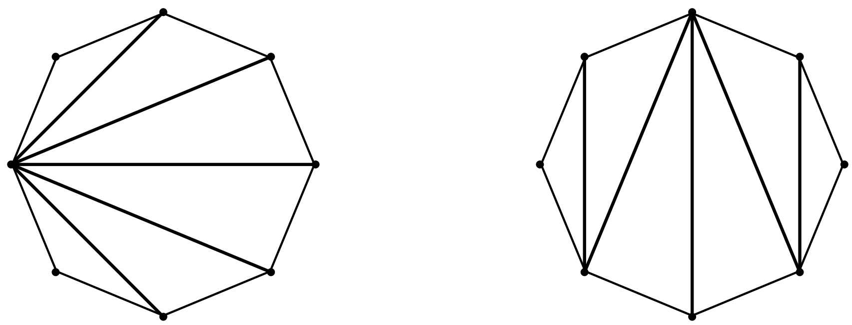


Figure 2: Two example triangulations of a regular octagon.

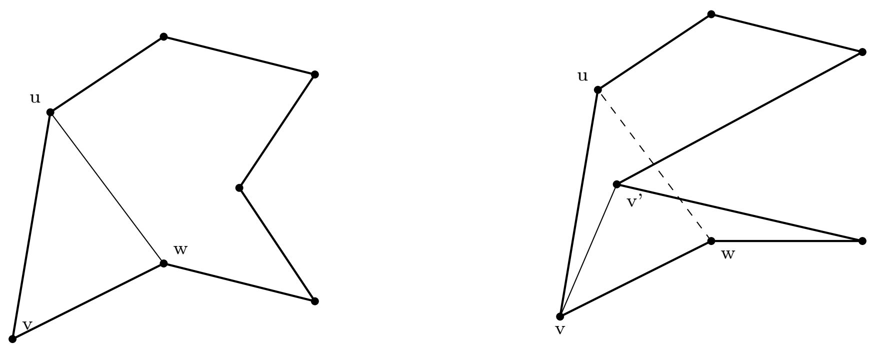


Figure 3: The two cases when trying to draw a diagonal between two points around a starting point.

polygon with m or fewer vertices can be triangulated and our goal is to show that that a polygon with n vertices with $n = m + 1 > 3$ can be triangulated.

I will begin by showing that a diagonal exists in the polygon, P, formed by n vertices. Take some vertex v from P and label the vertices neighboring it u and w. If u and w are connected and do not intersect with any edges of P, then a diagonal exists. Otherwise, there exists one ore more vertices of P that are inside triangle uvw. Now, imagine a line parallel to segment uw that goes through v. Let this line sweep toward segment uw until it encounters a vertex of P inside triangle uvw and call this vertex v'. The segment vv' must be a diagonal because there are no edges of P between v and v' by the definition of v'. So, a diagonal must exist in P.

The existence of this diagonal means that P can be broken into two polygons that share the diagonal as a common edge, with a total of $n + 2$ vertices. There are $n + 2$ because the n outer edges of the polygon remain part of either new polygon, and the diagonal contributes one edge to each of the newly created polygon, adding two edges. Since each new polygon must have at least 3 vertices, the other must have at most $n - 1 = m$ vertices. By the inductive hypothesis, each of these new polygons may be triangulated.

□

In the process of proving that it is always possible to triangulate any polygon, we have also generated a simple algorithm to triangulate any polygon. If we begin the same way as the proof, drawing a diagonal between some pair of vertices, two smaller shapes are created. Each of these shapes can undergo the same process and create two more shapes. This can be repeated recursively until all the shapes created by the splits are just triangles, so all the diagonals drawn must constitute a triangulation.

3.2 Triangulations of a Set of Points

In the previous section, I showed that any polygon can be triangulated. The method to do this started by drawing a single new edge to form a triangle using two preexisting edges. But what if there are no edges at all, and you just have a set of points? You could construct any number of polygons for a given set of points. To make this algorithmic, we must choose a specific one of these polygons to start to triangulate. To do this, we return to the familiar convex hull.

Definition 3. ***Triangulation of a set of Points*** *FILL OUT THIS DEFINITION*

After computing the convex hull, our situation appears almost like it did in the previous section, but there are still points inside that are not accounted for. To continue down this path, we sincerely hope that the following theorem is true:

Theorem 2. *Any set of points S has a triangulation, T, such that Hull(S)$\subset T$. — notation might not be quite right here, we want the edges of the hull to be a part of the edges of T*

Proof. We follow a similar method to the previous proof, using induction. Again, the base case is a set of 3 points, with the obvious triangulation of a triangle, which is also the convex hull of the points. Our hypothesis is that, assuming that a set of points S of size n has triangulation T with *Hull(S)*$\subset T$, some $S' \supset S$ of size $n+1$ has triangulation T' with *Hull(S')*$\subset T'$.

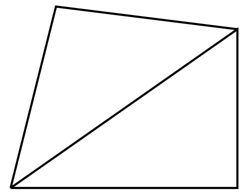
There are two cases for the new vertex v that belongs to S' but not S. Either v is inside *Hull(S)*, or it is outside of it. We will examine each case independently:

Explain that a point outside will have a new hull with two ”new edges”, and if these two edges are added to the previous triangulation, we still have a triangulation.

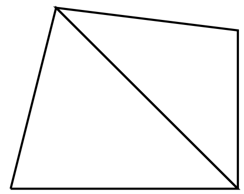
Explain how a point inside the hull is in a triangle, and hence this triangle can just have each vertex connect to the point inside and maintain a triangulation

...

□



(a) Non-Delaunay Triangulation



(b) Delaunay Triangulation

3.3 Delaunay Triangulations

Now that we know that we can generate some triangulation for any set of points, we can begin to look at a specific type of triangulation, the Delaunay triangulation. The Delaunay triangulation properties that give it more utility than other triangluations, and is the primary tool in generating the crust. This section is dedicated to defining it and exploring some interesting results to better understand why it is used to compute the crust.

Definition 4 (Delaunay Triangluation)**.** *The triangulation of a set of points S such that the circumcircle of each triangle only contains the vertices of that triangle, and no other points in S*

Something you may have noticed in the definition of a Delaunay triangulation is that I claimed that it is *the* triangulation that fits the criteria, not just some triangulation that does. This uniqueness is an important property of the Delaunay triangulation, but it is not at all apparent. Unfortunately this feature can be difficult to prove, and must be taken to be true for now. I will return to this discussion and provide an explanation of this will be provided in the section on Voronoi diagrams. Taking a step back, we should show that a Delaunay triangulation exists for any set of points, otherwise it would be useless in computational geometry.

Even before proving the existence of the Delaunay triangulation, I must build up to it with a few lemmata. I will begin by proposing a way to compare triangulations to one another on the basis of the interior angles of their triangles. Consider two triangulations T and T' of the same set of points S, each with n triangles. Each of these triangulations has a set $A = \{a_1, a_2, ..., a_{3n}\}$ of interior angles, ordered from smallest to largest. We say that $T > T'$ if A is lexicographically larger than A'.

Next, I will introduce three classifications of edges in a triangulation, 'thin', 'fat', and 'locked'. These classifications come into play when looking at the two triangles adjacent to an edge, making them a local

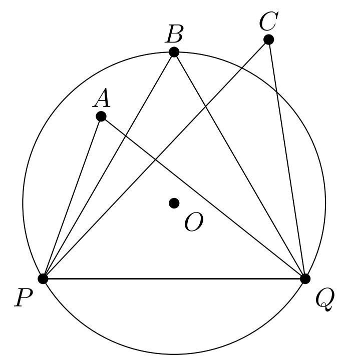


Figure 5: Three points with the same central angle inside, on, and outside the circle

feature of a triangulation, rather than a feature about the whole triangulation itself. Since these edges have a triangle on each side, they serve as a diagonal of a quadrilateral which I will refer to as the 'surrounding quadrilateral'. The main idea with these classifications is the relationship with the other diagonal created by the other pair of nonadjacent vertices. Changing the diagonal of a quadrilateral from one diagonal to the other is considered 'flipping,' which is an idea of great value in the study of triangluations.

Definition 5 (Thin, Fat, and Locked Edges of a Triangulation)**.** *A locked edge is an edge whose surrounding quadrilateral is nonconvex, and thus cannot be flipped without violating the definition of a triangulation of a polygon. If an edge is not locked, it is a part of some triangulation of T of its surrounding quadrilateral, with T' being the triangulation after flipping. An edge is thin if $T < T'$ and fat if $T > T'$*

Now, I will relate the definition of thin and thick edges to the empty circle definition of the Delaunay triangulation.

Lemma 1 (Thales')**.** *Consider a circle centered at point O concentric with points P, Q, and B. For some point A inside the circle and C outside the circle, both on the same side of P and Q as B, $\angle PCQ < \angle PBQ < \angle PCQ$. (Figure 5)*

Proof. It is well known that an inscribed angle is half the central angle. Consider a circle that again goes through P and Q, but now goes through C. This new circle will be centered at a different point, say O'. This radius of this new circle must be larger than the original since it connects a point outside the original. Because P, Q, and C are concentric, $\overline{PO'} = \overline{CO'} = \overline{QO'}$, but $\overline{PQ}$ remains the same as it was in the original circle. This means that the isoscles $\triangle PO'Q$ has its equal legs longer than $\triangle POQ$, which implies that the angle between these legs is smaller. This smaller angle is the central angle corresponding to the inscribed $\angle PCQ$. So since $\angle POQ > \angle PO'Q$, so too is $\angle PBQ > \angle PCQ$. The same argument follows for $\angle PAQ > \angle PBQ$. $\square$

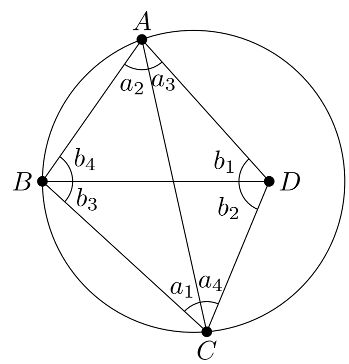


Figure 6: Angle labels used in Lemma 2

Lemma 2. *The triangles created by bisecting a quadrilateral are empty of the fourth vertex if and only if the edge bisecting them is thin.*

Proof. Consider a convex quadrilateral $ABCD$ with a circle connecting A, B, and C, with D inside the circle, and bisect $ABCD$ with its diagonals. Label the angles as shown in Figure 6. Since $\triangle ABD$ is inside the circumcircle, $\angle a_1 < \angle b_1$ by Thales' theorem. With similar reasoning plus the angle sum of a triangle, $\angle a_2 < \angle b_2, \angle a_3 < \angle b_3$, and $\angle a_4 < \angle b_4$. The 'a' angles correspond to $\overline{AC}$ and the 'b' angles to $\overline{BD}$. Since there is some 'b' greater than each 'a', the 'b' angles are lexicographically greater than the 'a' angles, hence $\overline{AC}$ is a thin edge.

To prove the reverse direction, one can take D is outside the circumcircle and follow the same line of reasoning to show that $\overline{AC}$ must be thick. □

Lemma 3. *The Delaunay triangulation contains no thin edges.*

Proof. Since all triangles in a Delaunay triangulation are empty of other points in the set, by the previous corollary, none of the edges may be thin. □

This theorem serves an alternate definition to the Delaunay triangulation. In some texts, this is the way they are defined, reaching the empty circle property from a series of results in the other direction. Either way, this result helps expose the property that the Delaunay triangulation is the triangulation that best reduces triangles which are very long and thin. Perhaps this property serves to make the Delaunay triangulation the 'natural' choice for many applications, ensuring that triangles connect to their "nearest" neighbors, rather than extending far across the shape.

With this new definition, we one step away from proving that the Delaunay triangulation must always exist. The idea behind this proof is that thin edges can be flipped to thick edges one by one until there are no more thin edges, which makes it a Delaunay triangulation by our new definition.

Theorem 3. *A Delaunay triangulation exists for any (finite) set of points S.*

Proof. Starting from any triangulation T, one may flip any thin edge to a thick edge to create T'. By the definition of thin and fat edges, a flip will cause the smallest angle in the triangles adjacent to the edge to strictly increase, while leaving all other angles the same. As a result, if we call the index of this original smallest angle i in the ordering of the angles in T, all angles before i in T' will remain the same, and the value at i will be strictly greater than before. This means that after flipping any thin edge to a thick edge to go from T to T', $T' > T$. Since there are only a finite number of triangulations of S and flipping thin edges to thick creates a strictly increasing sequence of triangulations, there must be some 'maximum' triangulation where this process must terminate. At this 'maximum' triangulation, there must be no thin edges, otherwise there would be some greater triangulation, and if there are no thin edges, the triangulation must be Delaunay. □

So, the existence of the Delaunay triangulation is finally guaranteed. But it is important to note that this proof still does not guarantee that we have *the* Delaunay triangulation, just that we have *one* of them. In the process of stepping between increasing triangulations, it would be reasonable to imagine flipping the 'wrong' edge, walking down a path to a local maximum different than if edges are flipped. It turns out that this is not the case, which is a fascinating convenient result when designing an algorithm to compute the Delaunay triangulation, for we can flip edges in any order and still eventually reach the Delaunay triangulation. If this were not the case, we would need to put much more care into the design of our algorithm.

It is of note that all the results in this section have dealt with points in *general position*. If there are four points that are concentric, then there cannot be a triangulation that fits the Delaunay criteria. So in this case, there is in some sense two valid triangulations that fit the Delaunay criteria which are call these degenerate. I will not deal with these here since in the general case, randomly sampled points have almost no chance of four concentric points.

4 Voronoi Diagrams

The third and final tool of interest for the crust is the Voronoi diagram. Similarly to the other tools introduced so far, the Voronoi diagram takes in a set of points S and returns a structure with useful properties. But instead of connecting the points in a specific way, the Voronoi diagram consists of lines that pass in between the points to divide the plane into *Voronoi regions*. Each of these regions contains a single $p \in S$, and is constructed such that any other point is in the region is closer to p than any other point in S. This definition is often much more intuitive than that of the Delaunay triangulation, and lends itself to a wide array of applications. For instance, to find the nearest post office for any given house on a map, the map could be separated into Voronoi regions where S are the locations of the post offices. Then, whichever region a house falls into is simply nearest to the corresponding post office. Voronoi diagrams come up in nature, too; if many structures began to grow from a starting point at the same rate until they collided, like crystals for example, the area of each crystal would be the Voronoi region of the starting point.

The following is a formal definition of the Voronoi region from Devadoss and O'Rourke.

Definition 6 (Voronoi Region)**.** *For a point p in set of points S, The Voronoi region of p, Vor(p), is such that*

$$\mathrm{Vor}(p) = \{x \in \mathbb{R} \mid ||x - p|| \leq ||x - q|| \ \forall q \in S\}$$

We can also define Voronoi regions in a different way. Looking at any other point q in S, the perpendicular bisector of $\overline{pq}$ divides the plane into two half-planes, one of which contains p, which we will call H. Vor(p) for this two-point case is just the half-plane described above, and it obviously is for any pair of points. Going further and considering a three-point system, there are two half-planes that contain p. Clearly anything not in one or both of the half-planes cannot be in Vor(p), or conversely, everything in their intersection must be in Vor(p). We can extend this idea to the following definition:

Definition 7 (Voronoi Region (Alternate Definition))**.** *Vor(p) is the intersection of all half-planes that contain p.*

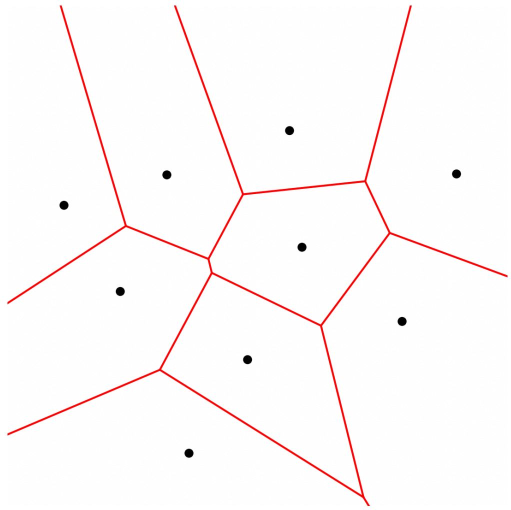
The benefit of this definition is that it clarifies a few interesting features of Voronoi regions. First, we can immediately see that Voronoi regions are convex by ... This follows from the INSERT THEOREM FROM CONVEX HULL SECTION. This definition is perhaps more indicative of what points in the plane are equidistant to multiple points in S: those that lie on the perpendicular bisectors that make up the edges of a Voronoi region. In fact, the points that are equidistant to multiple points in S are how the Voronoi diagram is defined:

Definition 8 (Voronoi Diagram)**.** *The Voronoi diagram of a set of points S is the collection of all points in the plane that are on the boundary of a Voronoi region.*

Or alternatively,

Definition 9 (Voronoi Diagram (Alternate Definition))**.** *The Voronoi diagram of a set of points S is the collection of all points in the plane that are closest to at least two points in S.*

It follows from this alternate definition that there is always a valid Voronoi diagram, and that there is only one such diagram for any S. This is because any point in the plane has an assignment: either it is



equidistant to multiple points in S or it isn't. Fortunately, this means there is much less work to do compared to the section on the Delaunay triangulation.

Let's turn our attention towards the edges and vertices of the Voronoi diagram. By its definition, any point in the Voronoi diagram is equidistant to at least two points in S. I will give a formal definition of Voronoi edges and vertices, but a visual example like Figure **??** is much easier to understand. The Voronoi edges are the straight lines in a Voronoi diagram and the vertices are the points where these edges meet.

The formal definitions for Voronoi edges and vertices is as follows

Definition 10 (Voronoi edges and vertices)**.** *A point in the plane belongs to a Voronoi edge if it is nearest to at least two points in S.*

A point in the plane is a Voronoi vertex if it is nearest to at least three points in S.

Each Voronoi vertex has a corresponding Voronoi disk with the vertex at its center:

Definition 11 (Voronoi Disk)**.** *A Voronoi disk of a set of points S is a maximal empty disk centered at a Voronoi vertex of S.*

These disks will always connect at least three points in S, because the definition of Voronoi vertex requires that there are at least three points equidistant. There must also exist no point $q \in S$ inside this circle, otherwise the vertex wouldn't belong to the Voronoi diagram because it would be closer to q than all other points in S. Voronoi disks will come into play in the section on the crust, but before that, they highlight a remarkable connection to the Delaunay triangulation. If we think about this disk a little further, we have a circle that connects three points in our set, and has no others inside, exactly the empty circle property that we used to define the Delaunay triangulation!

This turns out to be more than an interesting coincidence; it's actually a direct link between the Delaunay triangulation and the Voronoi diagram. If the Voronoi vertices are the centers of all circles that bound three points which have triangles obeying the empty circle property, then if we know all the Voronoi vertices, we also know which points to connect to create the Delaunay triangulation. This goes the other way, too. If we have the Delaunay triangulation, taking the circumcircle of each triangle returns all Voronoi vertices. And since the Voronoi vertices are where Voronoi edges meet, the centers of each circumcircle must be connected to form the Voronoi edges.

And from here we obtain our algorithm to compute the Voronoi diagram:

5 The Crust

It is finally time to put the geometric tools that this paper has focused on so heavily to use to define the crust of a set of points. As a reminder, the crust is a method of curve reconstruction introduced by [Amenta et al., 1998], which is where all the definitions come from in this section. It begins with a set of points taken from a smooth curve with the assumption that they are close enough together to be reconstruction. This ”close enough” condition is defined later in the section on sampling. The crust has the following simple definition:

Definition 12 (The Crust)**.** *Let S be a set of finite points in the plane, and let V be the vertices of the Voronoi diagram of S. Next let $S' = S \cup V$ and consider the Delaunay Triangulation of S'. Any edge of the Delaunay triangulation of S' that connects two points in S is an edge of the crust.*

INSERT A FIGURE IN THIS SECTION WITH AN EXAMPLE SIMILAR TO PAGE 3 OF THE PAPER

This is a fairly simple definition, and yet I claim that it can often successfully reconstruct a curve from a set of points in the plane. Before formally proving that this is true for a certain sampling condition, I will provide some intuition on why this is the case. To do this, I will give an alternate but equivalent definition of the crust that uses the familiar empty circle property:

Definition 13 (Alternate Definition of The Crust)**.** *Let S be a set of finite points in the plane, and let V be the vertices of the Voronoi diagram of S. An edge connecting two points in S belongs to the crust of S if there is a disk containing no other points in $S \cup V$.*

It also helps to note that Voronoi vertices are at the corners of regions of proximity, so they are closer to the points from S in adjacent Voronoi regions than to any other points in S. This means that points that

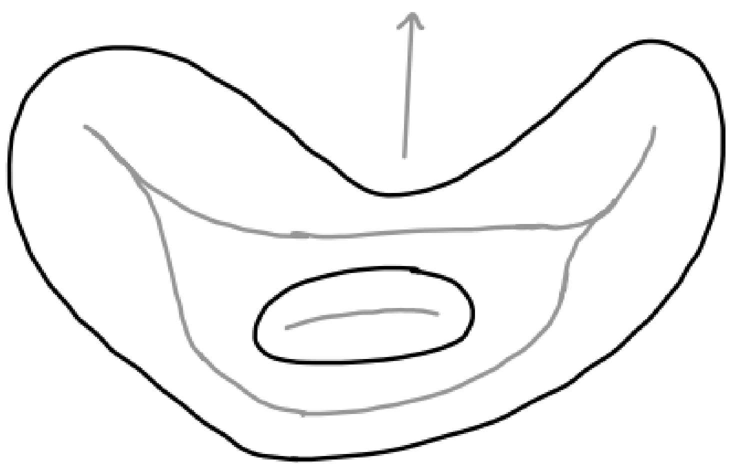


Figure 7: An example of the medial axis (gray lines) of a curve (black lines)

obey the alternative definition are connected with those in nearby proximity regions, not simply points that are closest together.

Also, looking at a few examples of Voronoi diagrams of many points, one can see that the diagram draws a line approximately through the middle of the shape that they create. This line is called the medial axis and will be explained further in the next section, but the idea is that the Voronoi edges trace out a line through the middle of a shape, then the empty circle properly serves to prevent lines from being drawn through this center line of the shape.

5.1 The Medial Axis

This section focuses on the medial axis, a tool that is used to define the conditions for sampling that guarantee that the crust can reconstruct a curve. It is essentially a Voronoi diagram for a continuous curve, rather than a discrete set of points. This becomes clear by the following definition that looks very similar to the alternate definition of the Voronoi diagram.

Definition 14 (The Medial Axis)**.** *The medial axis of a smooth curve F is the set of all points in the plane that are nearest to at least two points in F.*

An example is shown in Figure 7, where the medial axis appears to be a sort of midline of the curve.

The following lemma will get us lots of mileage when proving results about the medial axis.

Lemma 4. *Any disk B containing at least two points of a smooth curve F either intersects F in a single curve segment or contains a point of the medial axis (or both).*

Proof. If $B \cap F$ is a single curve segment, we are done.

If the intersection contains a closed loop, then there is a medial axis internal to the loop which is inside B, so we are done.

Otherwise, the intersection consists of two or more connected components F. Let c be the center of B and p be the nearest point of F to c. If there is more than one such p, c is a point of the medial axis, and we are done. If not, consider q, the nearest point to c on a different curve segment than p. Since c is the center and $q \in B$, any point on $\overline{cq}$ is closer to q than any point outside B. If we imagine walking along $\overline{cq}$, we start closest to p, and at some point cross over and are closer to q. At this point, we are nearest to both p and q, so this point belongs to the medial axis, and we are done. $\square$

And the next lemma allows us to connect Voronoi vertices to the medial axis, bridging our discrete geometric tools used to generate the crust with the continuous nature of a smooth curve and medial axis.

Lemma 5. *Any Voronoi disk B of a finite set $S \subset F$, where F is a smooth curve, must contain a point of the medial axis of F.*

Proof. There are two cases:

1. All samples are on the border of B and no part of F is inside. In this case, the center of B is a part of the medial axis.
2. In the neighborhood of some sample s, $G = F - s$ intersects B. Here, there are three more cases:
   - G lies entirely on the edge of B. Here, the center of B is part of the medial axis.
   - G is entirely inside B. Here, there exists a smaller disk B_ρ inside B that does not intersect G in a single curve segment, so lemma 4 applies
   - G crosses from outside to inside B at s, then it must leave elsewhere creating a curve segment. Along with the other samples on the edge of B, $F \cap B$ is not a single curve segment, so lemma 4 applies

$\square$

5.2 Sampling

This section I provide an empirical definition of sampling, basically meaning how close together points are taken from a smooth curve. Sampling is defined using a *local feature size* function at any given point. This function does what it says in the name, returning a number indicating how big features are at the point passed in. So, if there is a large flat region, the local feature size will be large, and if there are tight turns, then it will be small.

Definition 15 (Local Feature Size)**.** *The local feature size at some point p on a smooth curve F, denoted $LFS(p)$, is the Euclidean distance from p to the nearest point on the medial axis.*

If we think of the medial axis as a sort of midline, in a wide chasm, the medial axis will be very far from the edges, so the local feature size will be large. On the other hand, through a winding tunnel, the midline is very close to either edge, so the local feature size is very small.

Now we can define a sampling condition, which is just a proportionality factor of our local feature size.

Definition 16 (r-sampled Curve)**.** *A smooth curve F is r-sampled by a set of points S if every $p \in F$ is within distance $r \cdot LFS(p)$ of a sample $s \in S$.*

And now we reach our last couple definitions describing disks that and points lie between two sample points.

Definition 17 (Curve Voronoi Disk and Vertex)**.** *A curve Voronoi disk is a maximal disk, empty of sample points that is centered at a point on a smooth curve. The point on the curve is the curve Voronoi vertex.*

We are finally equipped to start proving that the crust really will reconstruct a curve under the proper conditions. Before beginning these proofs, we should note that The first step on this path is to prove that a curve Voronoi disk always intersects the curve in a single curve segment.

Lemma 6. *A curve Voronoi disk, B, intersects an r-sampled curve F in a single curve segment if $r \leq 1$.*

Proof. Consider the contrapositive: If B does not intersect F in a single curve segment then $r > 1$. If this assumption is true, there must a point of the medial axis inside B by lemma 4. In this case, the radius of B must be greater than $LFS(p)$. And since the maximum radius of a curve Voronoi disk is the distance from p to the nearest sample point, which is $r \cdot LFS(p)$, the only way for a curve Voronoi disk to have radius greater than the local feature size is if $r > 1$. □

And moreover,

Lemma 7. *Let F be an r-sampled curve with $r \leq 1$. There is a curve Voronoi disk connecting each pair of adjacent samples.*

Proof. Let s_1 and s_2 be adjacent samples along F. The curve segment between s_1 and s_2 intersects the bisector of $\overline{s_1 s_2}$ at least once. Let p be a curve Voronoi vertex at one of these intersections. If the corresponding curve Voronoi disk, B, has both samples on its boundary, we are done. Otherwise, there exists some sample s closer to p than both s_1 and s_2 on some other part of F. If this is true, then B does not intersect F in a single curve segment, thus violating lemma 6, so no such s exists. □

This happens to be very similar to the empty circle property used to define the Delaunay triangulation. So similar, in fact, that we can make the following statement:

Theorem 4. *Let F be an r-sampled smooth curve in the plane, $r < 1$. The Delaunay triangulation of the set S of samples contains an edge between every adjacent pair of samples.*

Proof. For any triangle in the Delaunay triangulation, you can resize the circumcircle so that it connects any pair of points in the triangle at all possible center points. Do this for every triangle, and you have a collection of all circles that connect two adjacent samples. Obviously, each curve Voronoi disk must be in this collection of circles. ☐

This theorem should provide some confidence that the crust can actually reconstruct a curve, but there is still much work to be done to prove this result.

5.3 Curvature

Next, we turn to getting a concrete lower bound for how large the angle between points on a curve can be given the sampling conditions. In other words, we will show that there is a limit to how much a smooth curve F can bend given r. The idea here is to find a disk that is tangent to F that does not have any points in its interior, then say that the maximum curvature is along its edge. One such circle tangent to any $p \in F$ is a circle of radius $LFS(p)$.

Lemma 8. *A disk tangent to a smooth curve F at a point $p \in F$ with radius at most $LFS(p)$ contains no points of F in its interior.*

Proof. The largest empty disk tangent to p has a center point m'. Since this disk is maximal, there must be at least one other point on its boundary. This means that there is more than one nearest point to m', so m' belongs to the medial axis. So, the radius of the disk is at most $LFS(p)$. ☐

Now let's view some properties for such a circle. In Figure 8, c is the center point, so label the radii LFS for the local feature size Next, label the distance from s to p as $LFS \cdot r$ to represent that this s is the closest point allowed to p given that the curve is r-sampled. Now note the following properties:

- The length of $\overline{sx}$ is $LFS \cdot \sin \gamma$
- The distance between s and p, $LFS \cdot r = 2LFS \cdot \sin(\frac{\gamma}{2})$, so $\gamma = 2 \arcsin(\frac{r}{2})$
- $\alpha = \frac{\gamma}{2} = \arcsin(\frac{r}{2})$
- The angle between L and p and $\overline{sp}$ is α

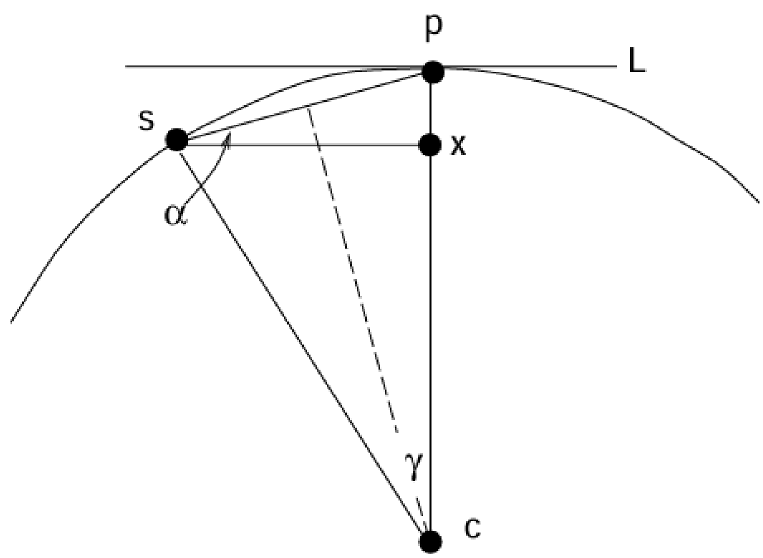


Figure 8: TEMPORARY – copied from paper, Line L is tangent to the circle at p, and the distance between c and p and c and s is L, and the distance from s to p is $L \cdot r$

Now we are ready to derive a lower bound on the curvature:

Lemma 9. *For an r-sampled curve, $r < 1$, the angle formed at a curve Voronoi vertex between two adjacent samples is at least $\pi - 2\arcsin(\frac{r}{2})$.*

Proof. By lemma 8, there can be no points of the curve inside a disk of radius $LFS(p)$ tangent to the curve. So, the tightest curve can be along the edge of the disk. In this case, the point labeled p in Figure 8 is in exactly this configuration. And from the figure it is clear that the triangle formed by p and two samples is isoscles with the duplicate angle equal to α. So, the angle formed at $p = \pi - 2\alpha = \pi - 2\arcsin(\frac{r}{2})$ by the properties listed above. □

And following closely from that,

Lemma 10. *For an r-sampled curve, $r < 1$, the angle spanned by three adjacent samples is at least $\pi - 2\arcsin(\frac{r}{2})$.*

Proof. Begin with the case of the curve Voronoi vertex with a corresponding central angle 2γ. The angle at p is an inscribed angle which is half of the central angle $2\pi - 2\gamma$. Now to add a third sample, just place another curve Voronoi vertex next to one of samples, then another sample next to that. Now there are two places with central angle 2γ, so the total central angle is doubled. This makes the corresponding inscribed angle, now at the middle sample, half of $2\pi - 4\gamma = \pi - 4\arcsin(\frac{r}{2})$ □

For lack of a better term, these results show that if you zoom in on any three samples from F, the curve appears pretty flat. As an example, for $r = 0.5$, the largest angle between any three sample points is just above $122°$.

5.4 Guaranteeing Reconstruction

Finally, we have the background to provide an upper bound on r so that an r-sampled curve will *always* be reconstructed by the crust. To prove that a curve is reconstructed, we need to prove that all adjacent edges are included in the crust and that no other edges are. I will provide this bound separately for each of these requirements, and the lower of the two will be an upper bound that guarantees the crust successfully reconstructs the curve. The idea behind each of these proofs is that I will be making the minimum assumptions required on r to give the mere possibility that the crust does not reconstruct the curve, then at the end I will flip the inequality on r, so this possibility does not exist.

Theorem 5. *The crust of an r-sampled curve with* $r < 0.40$ *contains an edge between every pair of adjacent vertices.*

Proof. An edge appears in the crust if and only if there is a circle connecting its vertices empty of other sample points and Voronoi vertices. By Lemma 7, we know that there is a curve Voronoi disk, B, touching each pair of adjacent vertices. So, B fits the condition to have its samples form an edge in the crust if it is also empty of Voronoi vertices.

So, I will provide a lower bound on r that *could* allow there to be a Voronoi vertex in B, then flip this bound to ensure it cannot exist.

Let the midpoint of B be called p. We know that the radius of B is at least $r \cdot LFS(p)$, by definition of r-sampled. Let Voronoi vertex v live inside B, with a corresponding Voronoi circle V with radius R, the distance to the nearest sample point on the boundary of B. Note that R is maximized when it is on the boundary of B, a distance $r \cdot LFS(p)$ from p. By lemma 5, V must contain some point of the medial axis.

Now, by the definition of local feature size, the disk, B', of radius $LFS(p)$ centered at p cannot contain a point in the medial axis. So, all there is to do is choose r so that some part of V is not inside B', so there is no guarantee that a point in the medial axis is inside B'. In other words, we need $r \cdot LFS(p) + R > LFS(p)$.

Let us now maximize R. We know that the angle between p and the two samples on the boundary of B is at least $\pi - 2\arcsin(\frac{r}{2})$ by the properties from Figure 8. Or on the other side, it is at most 2π minus this angle, $2\pi - (\pi - 2\arcsin(\frac{r}{2})) = \pi + 2\arcsin(\frac{r}{2})$. Let v be on this side so that it is as far as possible from the sample points. The furthest that v can be from either sample point is when it is at half this angle around the circle, at a maximum angle of $\frac{1}{2}(\pi + 2\arcsin(\frac{r}{2}))$, call this angle ψ. Clearly, $\frac{R}{2} \leq r \cdot LFS(p)\sin(\frac{\psi}{2})$, so finally:

$$r \cdot LFS(p) + 2r \cdot LFS(p) \sin\left(\frac{\pi + 2\arcsin(\frac{r}{2})}{4}\right) > LFS(p)$$

or

$$r + 2r \sin\left(\frac{\pi + 2\arcsin(\frac{r}{2})}{4}\right) > 1$$

It is easy to verify that the inequality is satisfied for $r > 0.40$, so we flip the inequality to guarantee that no such situation can occur to get $r \leq 0.40$ $\square$

Theorem 6. *The crust of an r-sampled smooth curve F does not contain any edge between non-adjacent vertices for $r <$ 0.252.*

Proof. We again want to make a situation where the assumption is not true, then flip the inequality at the end. So in this case, we begin by considering two nonadjacent vertices, s and t. We want there to exist some disk that connects s and t and contains either another sample or a Voronoi vertex. There are infinitely many disks that connect s and t, all with their center along the bisector of $\overline{st}$. We will limit this infinite continuum to have two extrema, disks V and V', with midpoints v and v'. These extrema are defined such they are as far out as possible while still satisfying that any disk with a center along $\overline{vv'}$ intersects at at least one of v and v'. With this construction, there is a disk that transitions from intersecting v' to intersecting v, at which point there exists a disk B that intersects s, t, v, and v'.

Without loss of generality, observe that if there is any sample s' in V, then there is a Voronoi vertex in B because a circle smaller than V that connects s, t, and s' with its center not at an endpoint of $\overline{vv'}$, and this center is equidistant to three samples, s, s', and t so it is a Voronoi vertex.

So again, our task is to provide the minimum conditions that such an s' can exist.

To do this, begin by focusing on the bigger of V and V', without loss of generality say V. Given that s' exists, $F \cap V$ is not a single curve segment, so by lemma 4, some point of the medial axis m must be located in V.

Let us now focus on the angles in the area near s. Because v and v' mark the diameter of B and s is a point on the exterior, they form a triangle inscribed in a semicircle. It is well known that in such a triangle, the inscribed angle is $\pi/2$. Since V and V' intersect at s, the angle between their tangent lines, ω, is vertical to this inscribed angle, and is thus also a right angle. Next, let us consider the smallest angle that s can make with its neighboring samples s_- and s_+. By lemma 10, $\angle s_- s s_+ \leq \pi - 2\arcsin(\frac{r}{2})$. This means that there is at least $\pi - 2\arcsin(\frac{r}{2}) - \omega = \frac{\pi}{2} - 2\arcsin(\frac{r}{2})$ left after subtracting away the angle from the tangent lines. Since ω is somewhat in the middle of $\angle s_- s s_+$, it leaves an angle on each side, the largest of which,

ψ, has $\psi = \frac{1}{2}(\frac{\pi}{2} - 2\arcsin(\frac{r}{2}))$. Consider a case where ψ is on the same side as V, and s_+ is the sample here and assume, without loss of generality, that V has radius 1. Because ψ is the angle between a tangent line and the distance to a sample, it is exactly α from Figure 8. From the aforementioned properties of this angle, we know that the distance between s and s_+ must be at least $2\sin(\frac{\pi}{2}) - 2\arcsin(\frac{r}{2})$.

And since the curve is r-sampled, this distance must be less than one half the sampling condition: $\sin(\frac{\pi}{2}) - 2\arcsin(\frac{r}{2}) \leq r \cdot LFS(p)$

Now, there must be a curve Voronoi vertex, p between s and s_+ by lemma 7. And recall we assumed V has radius 1 and contains a point m of the medial axis. This means that m is at most a distance 2 away, which gives means that $LFS(p) \leq 2$.

Finally, we get our inequality only in terms of r:

$$2\sin(\frac{\pi}{2}) - 2\arcsin(\frac{r}{2}) \leq 2r$$

$r > 0.252$ satisfies the inequality, which means that such an s' can exist inside V, and thus a Voronoi vertex inside B, allowing nonadjacent vertices to belong to the crust. If we flip the inequality, no such s' can exist, then $0 < r \leq 0.252$.

□

So that does it. We have shown that for $r \leq 0.252$, the crust will *always* reconstruct the curve correctly.

6 Examples

References

[Amenta et al., 1998] Amenta, N., Bern, M., and Eppstein, D. (1998). The crust and the -skeleton: Combinatorial curve reconstruction. *Graphical Models and Image Processing*, 60(2):125–135.

[Aurenhammer, 1991] Aurenhammer, F. (1991). Voronoi diagrams—a survey of a fundamental geometric data structure. *ACM Comput. Surv.*, 23(3):345–405.

[Devadoss and O'Rourke, 2011] Devadoss, S. L. and O'Rourke, J. (2011). *Discrete and computational geometry.* Princeton University Press, Princeton, NJ.